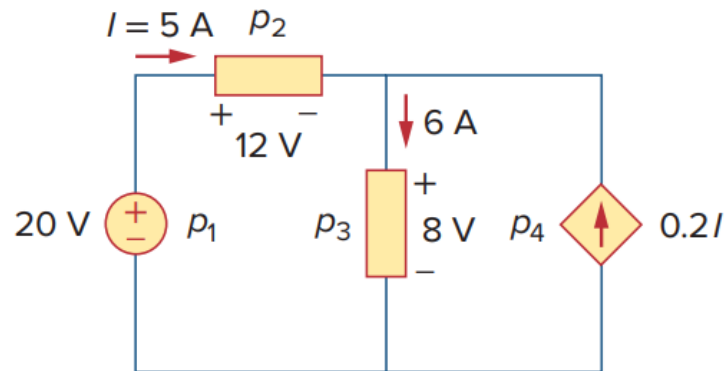
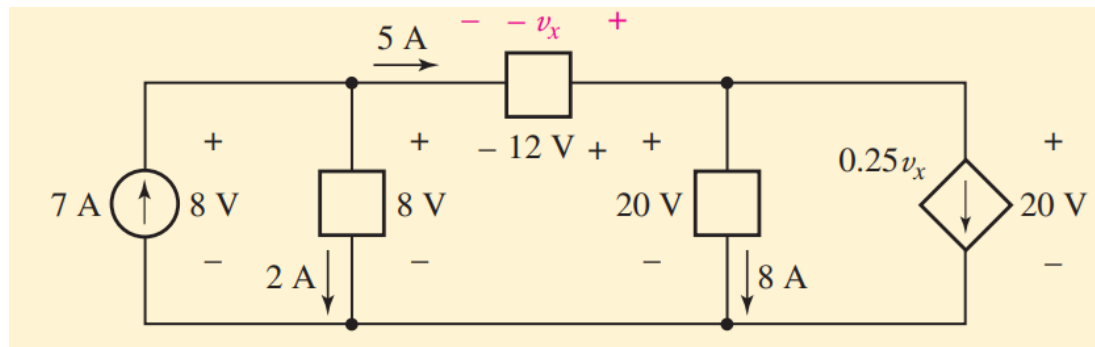




Find the power absorbed by each of the elements in

Resistance and Ohm's Law

In our story of voltage and current, we assumed that the movement of electrons through the conductor is straightforward...but in reality...free electrons collide with the ions, other electrons, atoms in the path of their motion. there is an **opposition** to the flow of charge in the conductor.

This **opposition** to the flow of charge through an electrical element is called **resistance**, and it's measurable which has the units of **ohms** and uses the Greek letter **omega** Ω as its symbol.

ideally, conductor are approximated to offer zero resistance!!

$$\text{Effect} = \frac{\text{cause}}{\text{opposition}}$$

$$\text{Current} = \frac{\text{potential difference}}{\text{resistance}}$$

$$I = \frac{E}{R} \quad (\text{amperes, A})$$

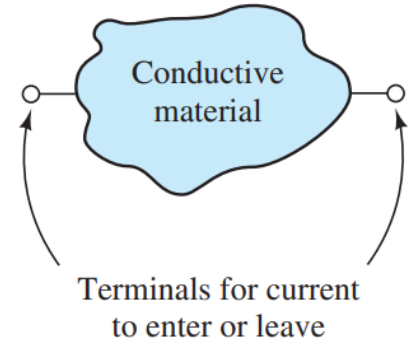
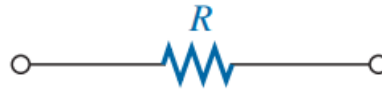
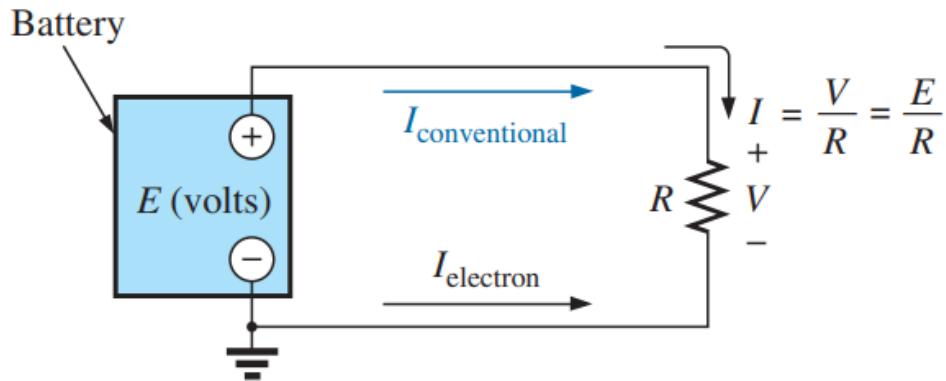
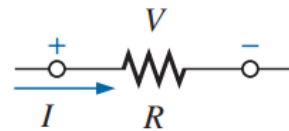


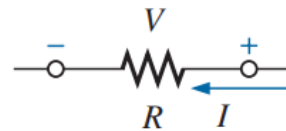
Figure 1.37 We construct resistors by attaching terminals to a piece of conductive material.



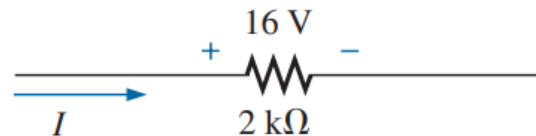
For any resistor, in any network, the direction of current through a resistor will define the polarity of the voltage drop across the resistor



(a)



(b)



Power supplied=Power absorbed.

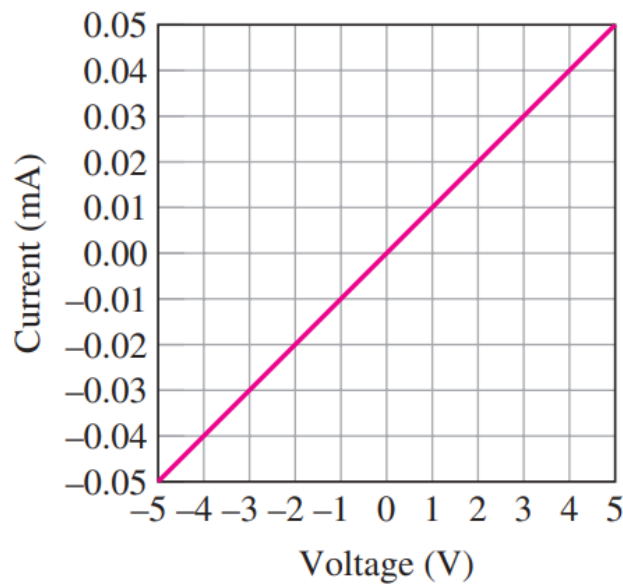
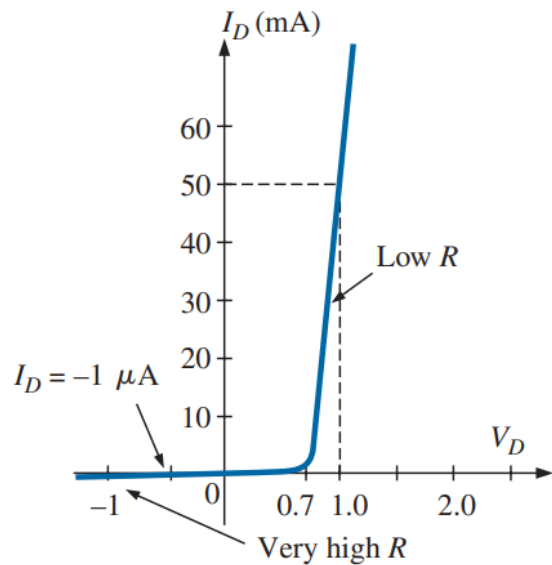
$E_i = V_i$, $E = V = iR$

Voltage rise (E) = Voltage drop (V)

$V = iR$, $V \propto I$

What determines the level of current that results when a particular voltage is applied?

Determine the magnitude of the current flowing through a $4.7 \text{ k}\Omega$ resistor if the voltage across it is (a) 1 mV ; (b) 10 V ; (c) $4e^{-t} \text{ V}$; (d) $100 \cos(5t) \text{ V}$; (e) -7 V .



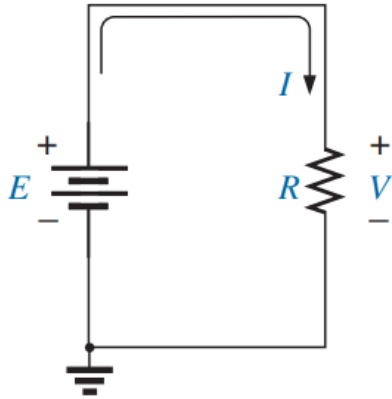
$$\frac{i}{v} = \frac{1}{R} = G$$

$$P = VI = V \frac{V}{R} = \frac{V^2}{R}$$

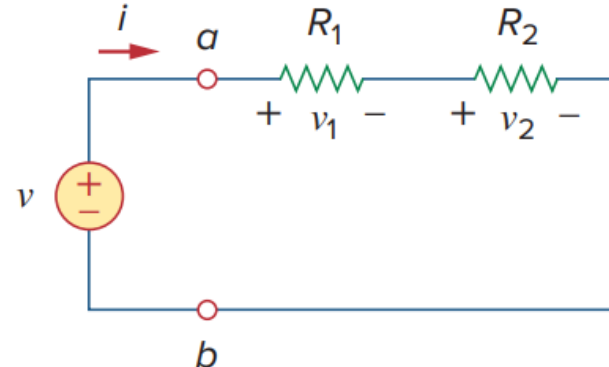
$$P = VI = IR \times I = I^2 R$$

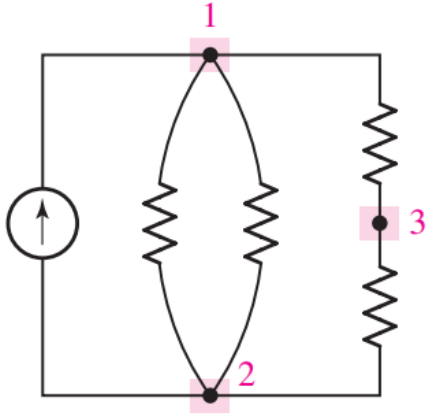
$$1 \text{ S} = 1 \text{ } \Omega^{-1} = 1 \text{ A/V}$$

$$p = vi = v^2 G = \frac{i^2}{G}$$



- For only two elements, Voltage source and a resistor, we can employ Ohm's law to determine current.
- We know voltage across the resistor, we know current!
- How can we determine voltage and current if there are more than one passive elements in the circuit connected in various arrangement?
- Circuits must be analyzed to determine a complete set of voltages and currents.
- In addition to Ohm's law, two more simple laws are needed.
- These new laws are **Kirchhoff's current law (KCL)** and **Kirchhoff's voltage law (KVL)**.





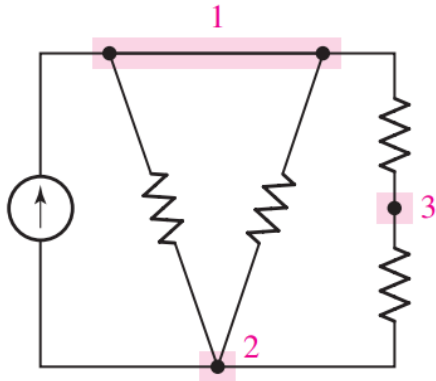
- A **node** in an electrical circuit is a point at which two or more circuit elements are joined together.
- If the point doesn't belong to a closed network (ckt), then it's not a node doesn't matter how many elements are connected to it!! (floating node, no current)
- Two nodes with no element in between are the same node.
- **Voltage value at any node or collection of same nodes is constant.**

If no node was encountered more than once as you move along the circuit, then the set of nodes and elements that we have passed through is defined as a **path**.

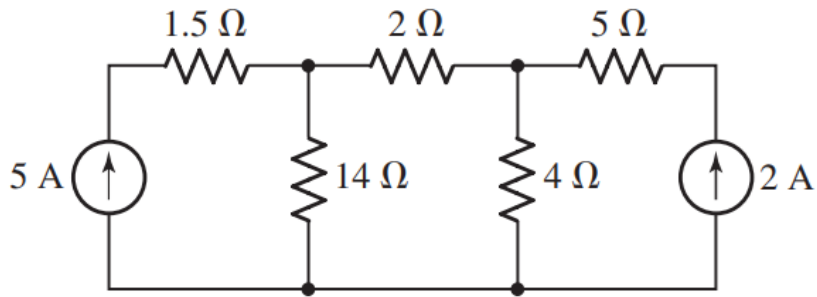
A **branch** represents a single element such as a voltage source or a resistor.

If the node at which we started is the same as the node on which we ended, then the path is, by definition, a closed path or a **loop**.

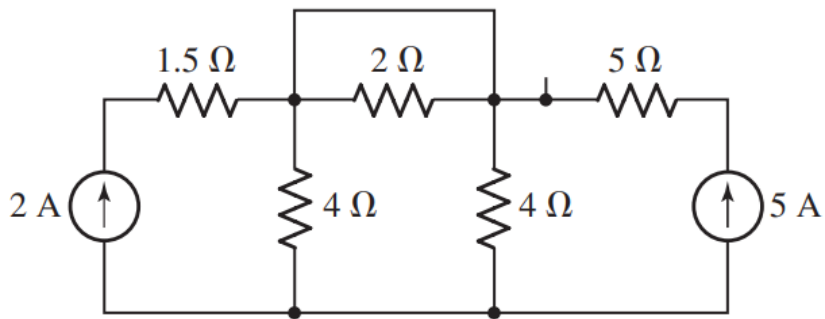
Independent loop: A loop is said to be independent if it contains at least one branch which is not a part of any other independent loop.



Referring to the circuit depicted in Fig. below, count the number of (a) nodes; (b) elements; (c) branches

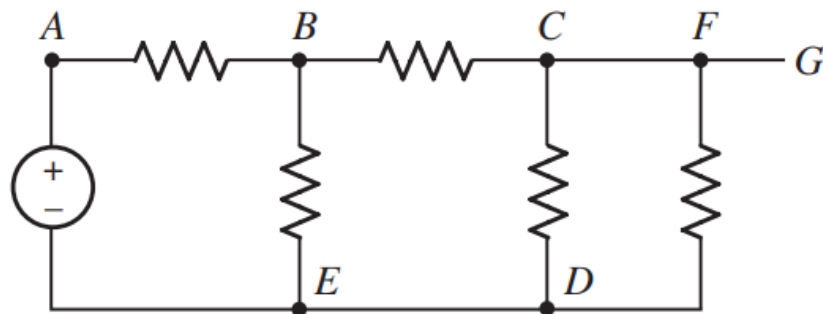


■ FIGURE 3.44



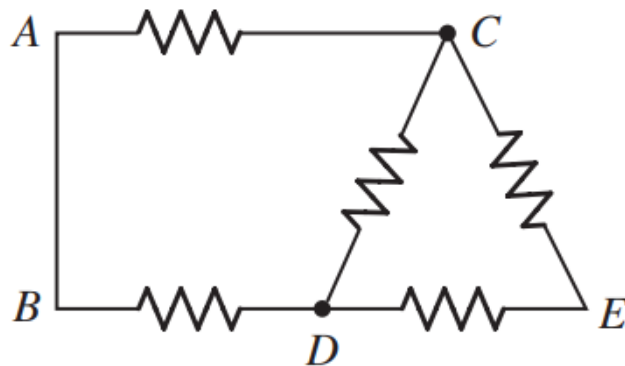
3. For the circuit of Fig. 3.46:

- (a) Count the number of nodes.
- (b) In moving from A to B , have we formed a path? Have we formed a loop?
- (c) In moving from C to F to G , have we formed a path? Have we formed a loop?



5. Refer to the circuit of Fig. 3.47, and answer the following:

- (a) How many distinct nodes are contained in the circuit?
- (b) How many elements are contained in the circuit?
- (c) How many branches does the circuit have?
- (d) Determine if each of the following represents a path, a loop, both, or neither:
 - (i) A to B
 - (ii) B to D to C to E
 - (iii) C to E to D to B to A to C
 - (iv) C to D to B to A to C to E



KCL: The algebraic sum of the currents entering any node is zero.

The sum of the currents entering a node equals the sum of the currents leaving a node.

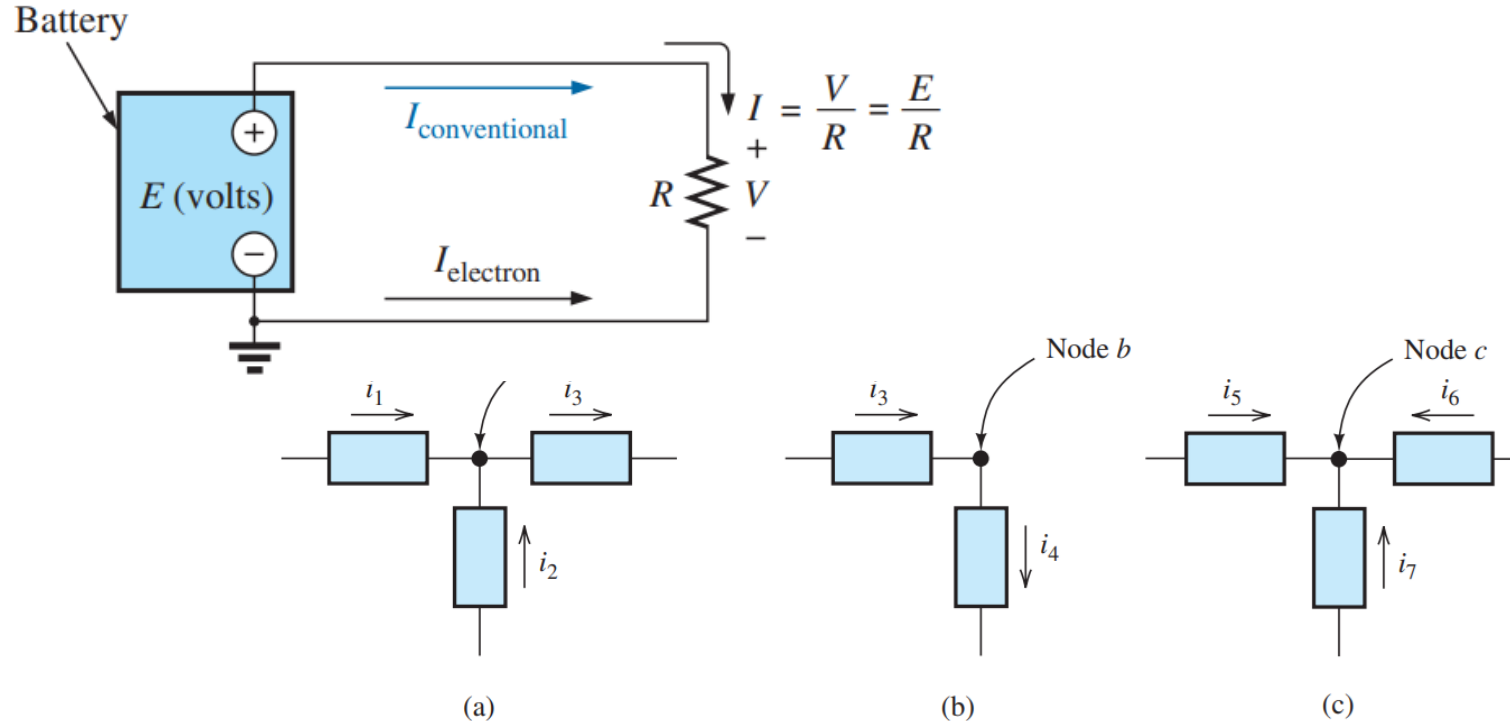
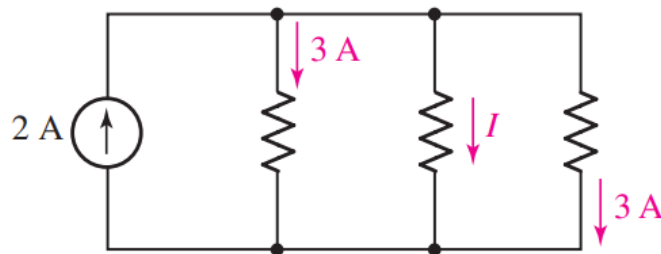
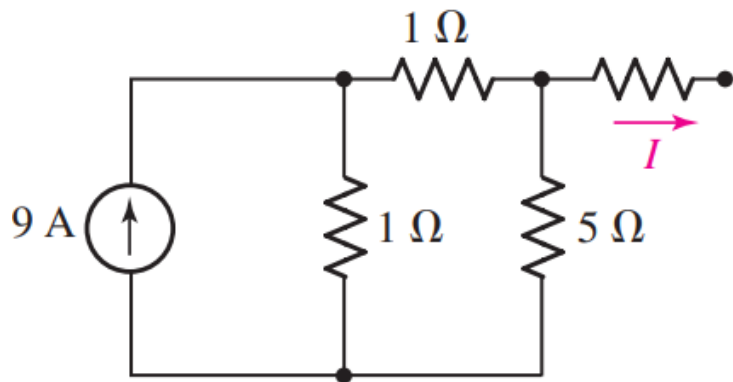
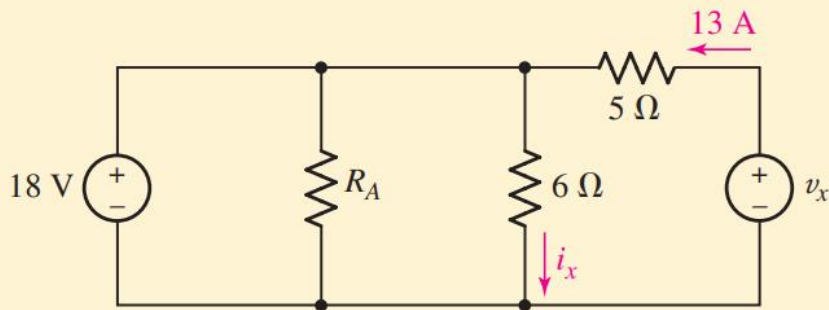


Figure 1.18 Partial circuits showing one node each to illustrate Kirchhoff's current law.



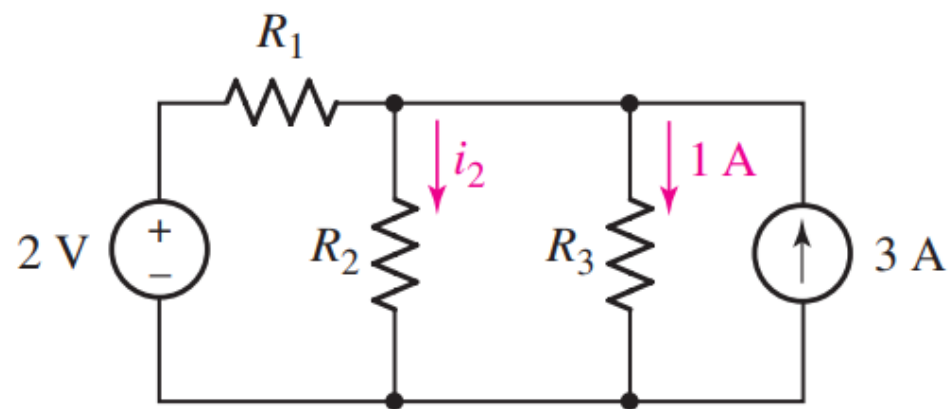
i_{R_1}

3.1 Count the number of branches and nodes in the circuit in Fig. 3.4. If $i_x = 3$ A and the 18 V source delivers 8 A of current, what is the value of R_A ? (Hint: You need Ohm's law as well as KCL.)

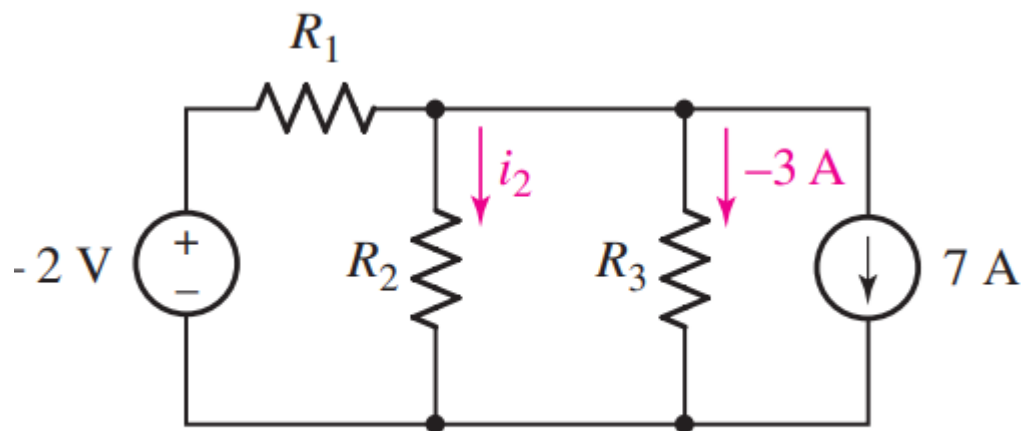


■ FIGURE 3.4

9. In the circuit shown in Fig. 3.51, the resistor values are unknown, but the 2 V source is known to be supplying a current of 7 A to the rest of the circuit. Calculate the current labeled i_2 .

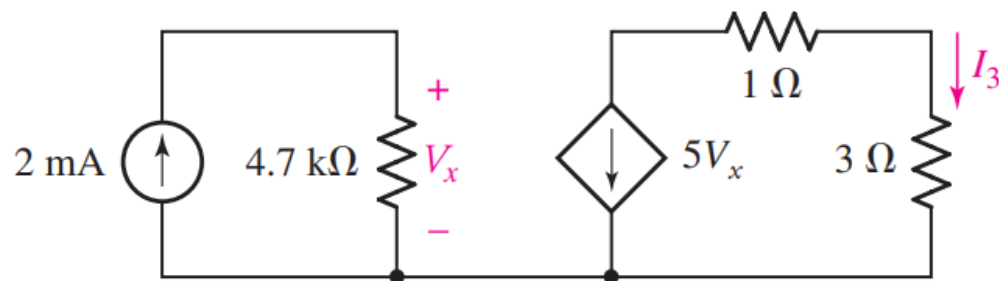


10. The voltage source in the circuit of Fig. 3.52 has a current of 1 A flowing out of its positive terminal into resistor R_1 . Calculate the current labeled i_2 .



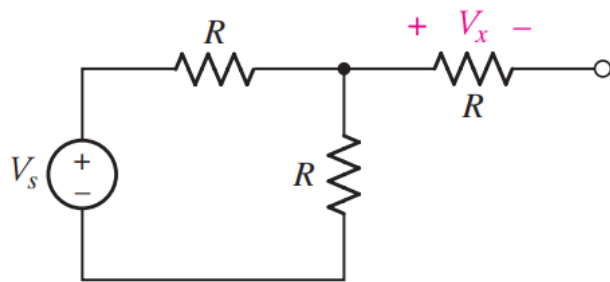
■ FIGURE 3.52

13. Determine the current labeled I_3 in the circuit of Fig. 3.55.



■ FIGURE 3.55

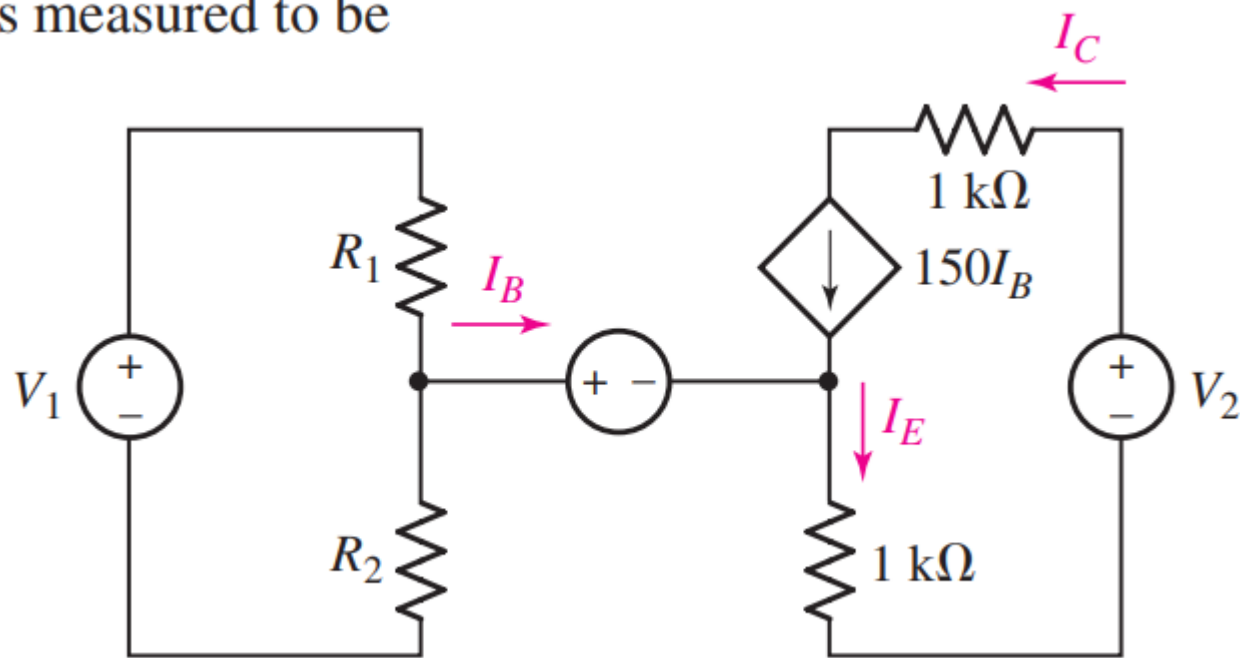
14. Study the circuit depicted in Fig. 3.56, and explain (in terms of KCL) why the voltage labeled V_x must be zero.



■ FIGURE 3.56

12. For the circuit of Fig. 3.54 (which is a model for the dc operation of a bipolar junction transistor biased in forward active region), I_B is measured to be $100\ \mu\text{A}$. Determine I_C and I_E .

s measured to be



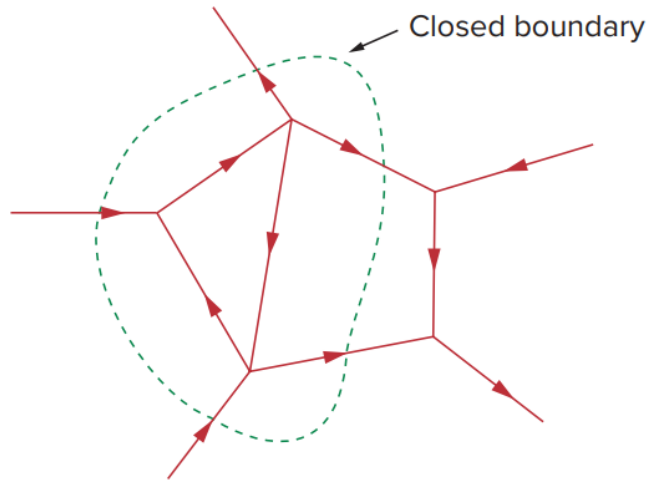


Figure 2.17

Applying KCL to a closed boundary.

total current entering the closed surface is equal to the total current leaving the surface.

